CSE476s - Fundamentals of Big-Data Analytics

Idea Selection Report

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Project Title

Big Data Analytics and Intelligent Insights from Amazon Consumer Reviews

Project Description

This project focuses on analyzing large-scale consumer review data from Amazon products using Big Data technologies and advanced analytics techniques.

The dataset, *Consumer Reviews of Amazon Products* (from Kaggle), contains thousands of customer reviews including ratings, review text, product information, and timestamps.

Using **Python**, **MongoDB**, and **Apache Spark**, the project processes and analyzes high volume unstructured textual data to extract meaningful insights.

Machine Learning (ML) and Deep Learning (DL) models are applied to perform sentiment analysis, rating prediction, and customer behavior modeling.

The goal is to transform raw customer feedback into actionable business intelligence that supports data-driven decision-making.

Technical Architecture

- **MongoDB**: Store large-scale semi-structured review data
- **Apache Spark**: Distributed big data processing and transformation
- **Python**: Data cleaning, preprocessing, visualization
- **Machine Learning**: Sentiment classification, rating prediction
- **Deep Learning (LSTM / RNN / BERT optional)**: Advanced text analysis

Business benefits

Analyzing Amazon consumer reviews is not just a technical exercise — it directly connects to how companies understand their customers and compete in the market.

Customer reviews represent raw, honest feedback. When this data is properly analyzed using Big Data tools like *Spark* and *MongoDB*, it becomes a powerful decision-making asset rather than just text stored in a database.

From a business perspective, this project helps companies clearly see what customers actually think about their products. Instead of relying only on star ratings, sentiment analysis allows deeper understanding of why customers are satisfied or dissatisfied. For example, a product may have a 4-star average, but text analysis might reveal repeated complaints about durability or packaging. This level of detail helps management prioritize improvements based on real evidence.

It also supports product strategy. By analyzing trends over time, companies can identify whether a new product launch is being positively received or if customer satisfaction is declining. Early detection of negative sentiment allows faster intervention before it affects sales or brand reputation.

On the market side, review analytics provides competitive insight. Companies can compare sentiment patterns across product categories to understand what features customers value most. This can guide marketing campaigns, pricing decisions, and product positioning.

Another important impact is customer retention. When patterns of dissatisfaction are identified early, businesses can adjust support strategies or improve product quality, reducing churn and strengthening brand loyalty.

From a strategic level, transforming large-scale review data into structured insights creates a data-driven culture. Instead of making assumptions, companies can rely on measurable indicators such as sentiment trends, review frequency, feature mentions, and rating prediction models.

In summary, this project turns unstructured customer feedback into actionable intelligence that improves product quality, marketing effectiveness, customer satisfaction, and overall market competitiveness.

Dataset Link

https://www.kaggle.com/datasets/datafiniti/consumer-reviews-of-amazon-products/data?select=Datafiniti_Amazon_Consumer_Reviews_of_Amazon_Products_May19.csv